

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Applied Physics

Semester: Spring

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Derive time period of torsion pendulum and find an expression for modulus of rigidity of the material of the suspension wire. 9

OR

Differentiate the terms free, damped and forced vibration. Develop the differential equation of a particle executing damped vibration in a medium. Explain the physical meaning of each term and each constant in the equations.

- b) Categorise the wave according to the modes of vibrations and show that the intensity of a progressive wave is directly proportional to the square of its amplitude. 6
2. a) Explain the construction and working of He-Ne laser with a suitable energy level diagram. Hence, explain its applications. 9
- b) The reverberation time of an empty hall is 1.5 sec. The same hall with 500 people is 1.4 sec. Find the reverberation time with 800 people in hall. 6
3. a) What do you mean by capacitance? Derive a relationship for charge stored at any time 't' in the capacitor in case of discharging and show that the charge decreases to 37% of its maximum value at capacitive time constant. 9
- b) A solid sphere of mass 40 kg and diameter 0.10 m is suspended on a wire. Find the period of angular oscillations for small displacement if the torque required to twist the wire is 4×10^{-3} Nm/rad. 6
4. a) Derive an expression for the resonant frequency in a forced EM oscillation. Hence, find the maximum current in the circuit. 9
- b) Show that $\nabla \cdot \vec{J} + \frac{d\rho}{dt} = 0$. Where symbols carry their usual meaning. 6

5. a) Using Schrodinger's wave equation obtain the energy and wave function of a small particle confined in an infinite potential well. And show that energy level are quantized. 9
- b) Show that, $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$, where $\vec{D}$ = displacement vector, $\vec{E}$ = electric field and $\vec{P}$ = polarizing vector. 6
6. a) Explain the modes of heat transfer with suitable examples. 9
- b) A refrigerator has to transfer an average of 200 J of heat per second from the temperature 15°C to 30°C . Calculate the average power consumed, assuming an ideal reversible cycle and no other losses. 6

OR

A thin metal plate is insulated on the back and exposed its front surface to solar radiation. The exposed surface of the plate has an emissivity of 0.7. If the solar radiation is including on the plate at the rate of 750 W/m^2 and the surrounding air temperature is 20°C , determine the surface temperature of the plate. Assume convection heat transfer coefficient $40 \text{ W/m}^2\text{K}$.

7. Write short notes on: (Any two) 2×5
- a) First law of thermodynamics
- b) Semi-conductor laser
- c) Inadequacy of classical mechanics

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Attempt all the questions.

1. a) Define SHM. Derive an expression for the time period of a physical pendulum and establish the interchangeability of centre of oscillation and suspension. 9
b) Calculate frequency of vibration of air particles in plane progressive wave of amplitude 2.18×10^{-10} m and intensity 10^{-10} W/m², the velocity of sound in air is 340 m/s and density of air is 0.00129 gm/cc. 6
2. a) What is interference? Explain the phenomenon of interference in thin film by reflected rays. 9
b) Obtain the charging time constant of a capacitor in a RC circuit such that charge through the resistor is decreased by 50% of its peak value in 5 seconds. 6
3. a) Draw appropriate circuit diagram for charging and discharging of a capacitor with Capacitance 'C' connected in series with a resistor of resistance 'R' and a battery of EMF 'E'. Starting with Kirchhoff's loop rule for charging circuit, derive charging equations. What is capacitive time constant of a charging circuit? 9
b) A string of mass 20 gm and length 1.6m is fixed on a rigid support at one end. The other end is attached with a mass of 2 kg. Calculate the speed of transverse pulse formed on the string. 6
4. a) State Biot-Savart's law. Use it to find the magnetic field intensity at any point due to a straight current carrying conductor. 9
b) Find the magnetic field intensity at the center of rectangular loop of length 'l' and width 'b' carrying current I. 6
5. a) What is LC oscillation? Discuss Oscillation of electric and magnetic energy in an LC circuit. 9
b) Explain briefly about the semiconductors doping and hence explain N-type and P-type conductors. 6

6. a) Explain physical significance of wave function. Discuss and find the energy of the particles travelling in one dimensional box of infinite height. 9
b) Prove that the electromagnetic waves travel with speed of light in vacuum. 6
7. Write short notes on: (Any two) 2×5
a) Reverberation and Reverberation time
b) Stimulated Emission
c) Faraday's Law of electromagnetic induction

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Applied Physics

Semester: Fall

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Attempt all the questions.

1. a) What is forced oscillation? Derive differential equation for forced oscillation and show that amplitude at resonance is inversely proportional to the damping constant of medium. 9

OR

What is a physical pendulum? Establish a formula for a physical pendulum's time period and show that the shortest duration is attained when the pendulum's length and radius of gyration are equal.

- b) The equation of transverse wave on a string is $y = (2.0\text{mm}) \sin [(20\text{m}^{-1})x - (600\text{S}^{-1})t]$. The tension on the string is 15 N. i) What is the wave speed ii) Find the linear density of the string in grams per meter. 6
2. a) Describe process of Laser emission and explain the construction and working principle of He-Ne laser. 9
- b) What is the reverberation time for a hall with length 10m, breadth 8m, and height 5m? If the coefficients of absorption of walls, ceiling and floor are 0.3, 0.4 and 0.5 respectively. 6
3. a) What is electronic polarization? Explain mathematically how relative permittivity is related with electronic polarizability using Clausius-Massotti equation. 9
- b) An optical fiber has core of RI 1.5 and cladding of RI 1.45. Calculate critical angle for core cladding interface, angle of acceptance in air and numerical aperture of the fiber. 6
4. a) Derive Maxwell's electromagnetic wave equation with their physical significance and express them in free space as well as in dielectric medium. 9
- b) Two capacitors of capacitances $25\mu\text{F}$ and $5\mu\text{F}$ are connected in parallel and charged with a 100V power supply. Calculate the total energy stored in the two capacitors. 6

5. a) Discuss the physical significance of the wave function and deduce the time dependent Schrödinger wave equation. 9
- b) A circuit has $L=5\text{ mH}$ and $C=2\text{ }\mu\text{F}$. How much resistance must be inserted in the circuit to reduce the resonant frequency by 5%? 6
6. a) Differentiate between heat engine and heat pump hence describe the COP of refrigerator with four essential components. 9
- b) Describe briefly about the mode of transfer of heat. 6

OR

Calculate the change in internal energy of 1 gm of water when boiled at $100\text{ }^{\circ}\text{C}$ at 1 atm pressure. The volume of steam at 100°C is 1671 cc and latent heat of vaporization is 540 cal/gm .

7. Write short notes on: (**Any two**) 2×5
- a) Energy and momentum operator
- b) Displacement Current
- c) Thermodynamic Process (Adiabatic and isothermal)

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Level: Bachelor
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Attempt all the questions.

1. a) What are the drawbacks of simple pendulum? Deduce expression for time period of compound pendulum, at which condition time period becomes minimum. 9
- b) A wave of frequency 300 cycles/sec has a phase velocity of 350 m/s. 6
 - i. How far apart are the two points which are 60° out of phase?
 - ii. what is the phase difference between two displacement of certain point at time 0.001 sec apart
2. a) Explain the necessary theory of the Newton's rings method for the determination of wave length of monochromatic light. Hence explain why the center of ring is dark in reflected case. 9
- b) A monochromatic light of wave length 5890 Å is incident normally on diffraction grating which has 6000 lines per centimeter a) At what angle will the second order image be seen? (b) Can you obtain the third order image with this grating? 6
3. a) What is electric dipole and dipole moment? Show that the electric field due to a short dipole along the axial line is twice the field along equatorial line of the dipole. 9
- b) A circuit contains 22kΩ resistor and fully charged 1000μF capacitor. What charge will remain on the capacitor after 15sec? 6
4. a) State Biot and Savart law and use it to find magnetic field strength on the axis of narrow circular coil. 9
- b) In Bohr's model of H-atom the electron circulation around the nucleus in path of radius 5.1×10^{-11} m at a frequency of 6.8×10^{15} rev/sec. Find out the value of 'B' at the center of the orbit. 6
5. a) Derive the Maxwell's electromagnetic wave equations in differential form with their physical significance. 9

- b) A 2μF capacitor is charged to 12V. The charging battery is then disconnected and a 10mH coil is connected across the capacitor so that LC oscillations occur. What is the maximum current in the coil? Assume the circuit contains no resistance. 9
6. a) What is wave function? Show that the difference in energy between any two consecutive energy levels of a free particle inside infinite potential well is inversely proportional to the square of well length. 6
- b) Entry of people in an auditorium of volume 160,000 cu ft and total absorption of 1000 sq ft of open window raises the absorption by 600 sq ft. Find change in reverberation time? 6
7. Write short notes on: (Any two) 2×5
 - a) Ruby laser
 - b) Self induction
 - c) Ponting vector

POKHARA UNIVERSITY

Level: Bachelor Semester : Spring Year : 2023
 Programme: BE Full Marks: 100
 Course: Physics Pass Marks: 45
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Attempt all the questions.

1. a) For a compound pendulum, define centre of suspension and centre of oscillation. Discuss the reciprocity of centre of suspension and centre of oscillation. 9
- b) A stretched string has a linear density 525 g/m and is under tension 45 N. We send a sinusoidal wave with a frequency 120 Hz and amplitude 8.5 mm along the string from one end. At what average rate does the wave transport energy? 6
2. a) What is diffraction of light? Show that width of central maxima is double than the other maxima in Fraunhofer's single slit. 9
- b) The radii of P^{th} and $(P+5)^{\text{th}}$ dark rings, in a Newton's ring experiment, are 0.6 Cm and 0.8 Cm. respectively. Calculate the wavelength of light used if the radius of the curved surface of the lens used be 100 Cm. 6
3. a) Define capacitance. Give a general method to calculate capacitance of capacitor. Apply this to find the capacitance of spherical capacitor. 9
- b) A room has dimensions $6 \times 4 \times 5 \text{ m}^3$. Find i) The mean free path of sound wave in the room. ii) Number of reflections made per sec by sound waves with the walls of the room. Given velocity of sound in air = 350 m/s. 6
4. a) State Ampere's theorem. Using Ampere's theorem, determine magnetic field due to a solenoid carrying current. 9
- b) In an RC circuit $E=20\text{V}$, $R=1.5\text{M}\Omega$ and $C=1.90\mu\text{F}$. calculate i) the time constant. ii) Maximum charge that will appear on the capacitor during charging iii) how long does it take for the charge to build up to $20\mu\text{C}$. 6

5. a) Write Maxwell's equation in integral form and differential form. Using Maxwell's equation in free space show that speed of electromagnetic wave is equal to the speed of light. 9
- b) In a Hall experiment, a current of 25A is passed through a long foil of silver, which is 0.1m thick and 3m long. Calculate Hall voltage produced across the width by a flux of 1.4 Weber/m^2 . Also, estimate the mobility and Hall coefficient. 6
6. a) What do you mean by wave function? Discuss its significance in quantum mechanics. Discuss the normalization condition and derive time independent Schrodinger wave equation. 9
- b) A circuit has $L=15\text{mH}$ and $C=10\mu\text{F}$. how much resistance should be added to circuit so that frequency of oscillation will 2% less than that of LC oscillation. 6
7. Write short notes on: (Any two) 2×5
 - a) Metal- semiconductor junction
 - b) Ruby LASER.
 - c) Hall Effect

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 Course: Applied Physics Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is simple harmonic motion? Derive an expression for the time period of a bar pendulum and show that the point of suspension and oscillation are interchangeable. 9

OR

What is torsional pendulum? Derive an expression for time period of torsional pendulum. Find the expression for modulus of rigidity of the wire.

- b) Find the expression for experimental value of g (acceleration due to gravity) by using time period of compound pendulum. 6
2. a) What is reverberation time? Find the expression for Sabine's Formula. What are its significance in our daily life? 9
- b) The size of an empty assembly of bell has dimensions $20 \times 15 \times 5 \text{ cm}^3$ and the reverberation time is 3.5 sec. What area of the wall should be covered by curtain cloth to reduce the reverberation time by 2.5 sec; if the absorption coefficient of the curtain cloth is 0.5. Also calculate the average absorption coefficient of the bell. 6
3. a) What is LASER? Explain the construction & working principle of He-Ne Laser with necessary energy level diagram. Also write the applications of LASER. 9
- b) What do you mean by acceptance angle & numerical aperture in optical fibre. An optical fiber has fractional refractive index change of 0.2 and a cladding refractive index 1.59. Determine acceptance angle for the fiber in water, which has refractive index of 1.33. Also evaluate Numerical Aperture (N.A.). 6
4. a) What is Capacitor? Discuss charging and discharging of capacitors. 9

- b) Obtain the time constant of a capacitor in RC circuit such that the current through the resistor increases to one third of its peak value in 5 seconds. 6

5. a) What is Maxwell Equation? Write Maxwell's equations in integral and differential forms? Convert the integral form of Maxwell's equation into its differential form. 9

- b) Using Maxwell's equation, Prove that $C = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$ Where symbols carry their usual meaning. 6

6. a) What is thermodynamic process? Differentiate between reversible and irreversible process. Find the expression for coefficient of performance of Refrigerator and Heat Engine. 9

- b) A brick wall 12 cm thick and 5 m^2 surface area is exposed to 250°C at one face and 50°C to another face. If the thermal Conductivity of the material is 1.5 W/mK , Determine the heat transfer rate. 6

OR

An ideal gas at 17°C has a pressure of 760 mm of Hg is compressed adiabatically until its volume is halved. Calculate the pressure and temperature of gas.

7. Write short notes on: (Any two) 2×5
- a) Time Independent SWE
- b) Progressive wave
- c) Clausius-Mossotti equation

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2022
 Programme: BE Full Marks: 100
 Course: Applied Physics Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) What is damped harmonic oscillation? Write down the equation for damped oscillation. Using the equation, explain what are underdamped, overdamped and critically damped oscillations. 10

OR

What is a compound pendulum? Show that there are four collinear points in a bar pendulum where the time period of oscillation is same. Also find out the situation when the time period will be minimum.

- b) A wire has a torsional constant of 2N.m/radian. A disk of radius 5 cm and mass 100 gram is suspended through the wire. Find the frequency of such torsional oscillations. 5
2. a) What is a wave? Starting with the relation for displacement of a particle, find the relation for particle velocity and particle acceleration. 5
- b) Does a plane progressive wave carry energy? Show that the intensity of a wave is directly proportional to the square of amplitude. 5
- c) A string of mass 20 gram and length 1.6 m is fixed on a rigid support at one end. The other end is attached with a mass of 2 kg. Calculate the speed of transverse pulse formed on the string. 5
3. a) What is a capacitor? Develop a relation for charge stored at any time 't' in the capacitor. Show that the current decreases to 37 percent of its maximum value at capacitive time constant. 10
- b) Show the relation: $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$ where the symbols carry their usual meanings. 5
4. a) Write down Maxwell's equation in integral form. Convert them into differential form. Explain the physical significance of each equation. 10

- b) There are two capacitors of capacitance 5 μF and 2 μF . These capacitors are connected in series. The combination is connected to an inductor of inductance 10 mH. Calculate the resonant frequency of such combination. 5

5. a) Explain the inadequacy of classical mechanics for the development of quantum mechanics. Also write down the importance of quantum mechanics. 5
- b) Write down the relation of the wave function. Use this to derive the Schrodinger time independent wave equation. 5
- c) Calculate the de-Broglie wavelength of a proton produced by applying 15 kV energy. (Take mass of proton = 1.67×10^{-27} kg). 5
6. a) Explain what are reversible and irreversible process in thermodynamics. Why in a gas system there are two types of specific heat capacities? Show that specific heat capacity of a gas at constant pressure is always greater than that at constant volume. 10

OR

What is emissive power? Apply this power to define blackbody and perfect black body. Explain Stefan-Boltzmann Law of blackbody radiation.

- b) A closed metal vessel contains water at 30°C. The vessel has surface area 0.5 m² and thickness of 4 mm. Calculate the heat loss per minute by conduction if the outside temperature is 15°C and thermal conductivity of metal is 400 Wm⁻¹K⁻¹. 5
7. Write short notes on: (**Any two**) 2×5
- a) Reverberation
- b) Laser
- c) Clausius - Mossotti equation

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Course: Physics

Semester: Spring

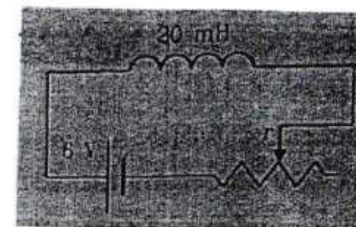
Year : 2021
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Attempt all the questions.

1. a) Why compound pendulum is preferred than simple pendulum? Prove that the minimum time period is obtained if the point of suspension and point of oscillation are equidistant from the center of gravity. 2+7
b) A source of sound has frequency of 256 Hz and amplitude of 0.50 cm. Calculate the energy flown across a square cm per second. Take the velocity of sound in air as 330 m/s and density of air as 1.29 kg/m^3 . 6
2. a) What is interference and why coherent sources are necessary for interference? Explain the necessary theory for Newton's rings method to determine the wavelength of monochromatic light. 2+7
b) Determine the specific rotation of the given sample of sugar solution if the plane of polarization is turned through 13.2° . The length of the tube containing 10% sugar solution is 20 cm. 6
3. a) Discuss about the advantages and applications of optical fiber in data transmission and hence explain the working principle of optical fiber. 2+7
b) Two charges $10 \mu\text{C}$ and $-10 \mu\text{C}$ are placed at points A and B separated by a distance of 10 cm. Find the electric field and potential at a point 'P' on the perpendicular bisector of AB at a distance 12 cm from its middle point. 6
4. a) What is electric dipole? Find out the potential and field due to an electric dipole at a distance 'r' from its center, which is lying along equatorial line. 1+8
b) Consider the circuit as shown in figure. The sliding contact is being pulled towards right so that the resistance in the circuit is increasing. Its value at the instant shown is 12Ω . Will the current be more than 0.50 A or less than it at this instant? 6



5. a) State Biot and Savart's law. Use it to find the magnetic field due to an infinitely long straight current carrying conductor. 1+8
b) Derive the time independent Schrodinger wave equation for a free particle. 6
6. a) Differentiate between the LC circuit and LCR circuit. Discuss the theory of LC oscillation. Hence prove that the frequency of LC oscillation is $f = \frac{1}{2\pi\sqrt{LC}}$. Also discuss the energy conservation in LC circuit. 9
b) Prove that $\vec{\nabla} \times \vec{B} = \mu(\vec{J} + \epsilon \frac{d\vec{E}}{dt})$ and write its physical significance. 6
7. Write short notes on: (Any two) 2×5
 - a) Describe any one method for producing ultra sound
 - b) Band theory of solids
 - c) Lorentz force